

Introduction to Derivatives (Optional)

This optional section introduces the mathematical concept of a **derivative**, which is the core component of **backpropagation**. Backpropagation is the algorithm that TensorFlow uses to automatically calculate the derivatives needed for gradient descent (or the Adam optimizer).

This lesson uses an intuitive, non-calculus-based approach to understand what a derivative is.

An Intuitive Definition of a Derivative

A derivative measures the **sensitivity** or **effect** that a tiny change in one variable has on another.

For a cost function $J(w)$, the derivative of J with respect to w (written as $\frac{\partial J}{\partial w}$) answers the question: "If I nudge w by a tiny amount, how much does J change in response?"

The "Epsilon Nudge" Method

Let's use a simple cost function: $J(w) = w^2$

1. **Start with a value:** Let $w = 3$.
 - The cost $J(3) = 3^2 = 9$.
 2. **Nudge the variable:** Increase w by a tiny amount, ϵ (epsilon). Let's say $\epsilon = 0.001$.
 - Our new w is $3 + 0.001 = 3.001$.
 3. **Calculate the new cost:**
 - $J(3.001) = (3.001)^2 = 9.006001$.
 4. **Analyze the change:**
 - We increased w by 0.001 (ϵ).
 - The cost J increased by 0.006001.
 - This change in J (0.006001) is almost exactly **6** times the change in w (0.001).
 5. **Conclusion:** The **derivative is 6**.
 - **Informal Definition:** If w goes up by ϵ , and $J(w)$ goes up by $k \times \epsilon$, then the derivative is k .
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How Derivatives Power Gradient Descent

The gradient descent update rule is: $w := w - \alpha \frac{\partial J}{\partial w}$

The derivative ($\frac{\partial J}{\partial w}$) tells the algorithm how big of a step to take.

- **Large Derivative:** If the derivative is large (like 6), it means changing w has a **big effect** on the cost J . Therefore, gradient descent should take a **large step** to get a big improvement.
 - **Small Derivative:** If the derivative is small, changing w has a **small effect** on J . Therefore, gradient descent should take a **small step**.
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Key Properties of Derivatives

1. The Derivative Depends on the Value of w

The derivative is not a fixed number; it changes depending on the *current value* of w . For our function $J(w) = w^2$:

- If $w = 3$, $\frac{\partial J}{\partial w} = 6$.
- If $w = 2$, $\frac{\partial J}{\partial w} = 4$. (A 0.001 nudge makes J go from 4 to ~ 4.004).
- If $w = -3$, $\frac{\partial J}{\partial w} = -6$. (A +0.001 nudge to w makes J decrease by 0.006, so the change is $-6 \times \epsilon$).

2. The Calculus Connection (Optional)

- **Slope:** In calculus, the derivative is formally defined as the **slope of the line tangent** to the function at that exact point.
- **Formula:** For $J(w) = w^2$, the derivative formula is $\frac{\partial J}{\partial w} = 2w$.
- This formula perfectly matches our intuitive findings:
 - At $w = 3$, derivative = $2 \times 3 = 6$.
 - At $w = 2$, derivative = $2 \times 2 = 4$.
 - At $w = -3$, derivative = $2 \times (-3) = -6$.

3. Examples of Other Derivatives

The video uses the Python library `SymPy` to find the derivative formulas for other functions:

Function ()	Derivative ()	Value at
w^2	$2w$	4
w^3	$3w^2$	12
w	1	1

Function ($J(w)$)	Derivative ($\frac{\partial J}{\partial w}$)	Value at $w = 2$
$1/w$	$-1/w^2$	-0.25

4. A Note on Notation

- You may see two symbols for derivatives: d/dw (total derivative) and $\partial/\partial w$ (partial derivative).
- The "**squiggly d**" (∂) is used when a function (like our cost J) depends on *more than one variable* (e.g., $w_1, w_2, \dots, b_1, b_2, \dots$).
- Since our cost function *always* depends on many parameters, this course will **always use the partial derivative symbol** $\partial/\partial w$.